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UNIT-IV (DOUBLE INTEGRALS)

LECTURE NO.:- 27

Double Integrals (Cartesian)

A double integral is something of the form $\iint_R f(x,y)dxdy$,

where R is called the region of integration and is a region in the xy-plane. The double integral gives us the volume under the surface $z = f(x,y)$ just as a single integral gives the area under a curve.

Let us consider an integral I = $\int_{x_1}^{x_2}\int_{y_1}^{y_2} f(x,y)dxdy$. The evaluation of this integral depends on the nature of the limits x_1, x_2, y_1 and y_2. So we may have the following cases:

1. **When all the limits are constants:** If x_1, x_2, y_1 and y_2 are constants, then the integration is possible in both orders; first w.r.t. x, then w.r.t. y and vice versa.
2. **When x_1, x_2 are constants and either y_1 or y_2 or both are functions of x:** In this case, $f(x,y)$ is first integrated with respect to y between the limits y_1, y_2 treating x as a constant and then the resulting expression is integrated with respect to x within the limits x_1, x_2, i.e.
$$\text{I} = \int_{x_1}^{x_2}\left\{\int_{y_1}^{y_2} f(x,y)dy\right\}dx.$$
3. **When y_1, y_2 are constants and either x_1 or x_2 or both are functions of y:** In this case, $f(x,y)$ is first integrated with respect to x between the limits x_1, x_2 treating y as a constant and then the resulting expression is integrated with respect to y within the limits y_1, y_2, i.e.
$$\text{I} = \int_{y_1}^{y_2}\left\{\int_{x_1}^{x_2} f(x,y)dx\right\}dy.$$

Examples of Double Integrals

(a) If $f(x,y) = 1$, then $\iint_R dxdy =$ Area of the region R.

(b) If $f(x,y) = z$, then $\iint_R zdxdy =$ Volume under the surface $z = f(x,y)$.

(c) If $f(x,y) =$ force per unit area on the plate, then $\iint_R f(x,y)dxdy =$ Force on the plate.

Properties of Double Integrals

(i) If the region R is partitioned into two parts, say R_1 and R_2, then
$$\iint_R f(x,y)dxdy = \iint_{R_1} f(x,y)dxdy + \iint_{R_2} f(x,y)dxdy.$$
Similarly for a subdivision of R into three or more parts.

(ii) The double integral of the algebraic sum of a fixed number of functions is equal to the algebraic sum of the double integrals taken for each term, i.e.

$$\iint_R [f_1(x,y)+f_2(x,y)+f_3(x,y)+...........+f_n(x,y)]dxdy$$
$$=\iint_R f_1(x,y)dxdy+\iint_R f_2(x,y)dxdy+\iint_R f_3(x,y)dxdy+.........+\iint_R f_n(x,y)dxdy.$$

(iii) A constant factor may be taken outside the integral sign. Thus

$\iint_R mf(x,y)dxdy = m\iint_R f(x,y)dxdy$, where m is a constant.

Example 1. Evaluate: $\iint_R ydxdy$ over the region R bounded by the parabolas $y^2=4x$ and $x^2=4y$.

Solution. The points of intersection of parabolas are given by $y^2=4x=4\sqrt{4y}$ as $x^2=4y$.

$\Rightarrow y^4=64y \Rightarrow y(y^3-64)=0 \Rightarrow y=0,4 \Rightarrow x=\sqrt{4y}=0, 4.$

So the two parabolas intersect at (0, 0) and (4, 4).

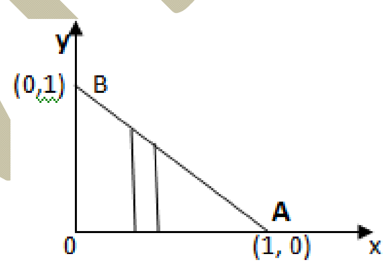
Taking vertical strip, we see that y varies from $\left(\frac{x^2}{4}\right)$ to $2\sqrt{x}$ along the strip and x varies from 0 to 4. So we have

$$\iint_R ydxdy=\int_0^4\int_{x^2/4}^{2\sqrt{x}} ydxdy=\int_0^4\left[\frac{y^2}{2}\right]_{x^2/4}^{2\sqrt{x}}dx=\int_0^4\left(2x-\frac{x^4}{32}\right)dx=\left[x^2-\frac{x^5}{160}\right]_0^4=16-\frac{4^5}{160}=16-\frac{16\times 64}{160}$$

$\Rightarrow \iint_R ydxdy=\frac{48}{5}$. Ans.

Example 2. Evaluate $\iint_R xydxdy$ over the region R in the positive quadrant for which $x+y\le 1$.

Solution. Let $I=\iint_R xydxdy$. Here $0\le x<1$ & $0\le y<(1-x)$.



$$\therefore I=\int_0^1\int_0^{1-x} xydxdy=\int_0^1 x\left\{\int_0^{1-x} ydy\right\}dx=\int_0^1 x\left[\frac{y^2}{2}\right]_0^{1-x}dx=\frac{1}{2}\int_0^1 x(1-x)^2dx=\frac{1}{2}\int_0^1(x+x^3-2x^2)dx$$

$$=\frac{1}{2}\left\{\frac{x^2}{2}+\frac{x^4}{4}-\frac{2x^3}{3}\right\}_0^1=\frac{1}{2}\left\{\frac{1}{2}+\frac{1}{4}-\frac{2}{3}\right\}=\frac{1}{2}\left\{\frac{6+3-8}{12}\right\}=\frac{1}{2}\times\frac{1}{12}=\frac{1}{24}.$$

$\Rightarrow \iint_R xydxdy=\frac{1}{24}$. Ans.

Example 3. Evaluate: $\int_0^1\left\{\int_0^1 \frac{x-y}{(x+y)^3}dy\right\}dx$.

Solution. Since $\frac{x-y}{(x+y)^3}=\frac{x-(y+x-x)}{(x+y)^3}=\frac{2x-(x+y)}{(x+y)^3}=\frac{2x}{(x+y)^3}-\frac{1}{(x+y)^2}$, so we have

$$\int_0^1\left\{\int_0^1 \frac{x-y}{(x+y)^3}dy\right\}dx=\int_0^1\left[\int_0^1\left\{\frac{2x}{(x+y)^3}-\frac{1}{(x+y)^2}\right\}dy\right]dx=\int_0^1\left[2x\int_0^1\frac{dy}{(x+y)^3}-\int_0^1\frac{dy}{(x+y)^2}\right]dx$$

$$=\int_0^1\left[2x\frac{1}{-2(x+y)^2}+\frac{1}{x+y}\right]_0^1 dx=\int_0^1\left[-\frac{x}{(x+1)^2}+\frac{1}{x+1}+\frac{x}{x^2}-\frac{1}{x}\right]dx$$

$$=\int_0^1\left[\frac{-x+(x+1)}{(x+1)^2}\right]dx=\int_0^1\frac{1}{(x+1)^2}dx=\left[\frac{-1}{(x+1)}\right]_0^1=-\frac{1}{2}+1=\frac{1}{2}.$$

$\therefore\ \int_0^1\left\{\int_0^1 \frac{x-y}{(x+y)^3}dy\right\}dx=\frac{1}{2}$. Ans.

Example 4. By evaluating the following integrals, prove that

$$\int_0^1\left\{\int_0^1 \frac{x-y}{(x+y)^3}dy\right\}dx\neq\int_0^1\left\{\int_0^1 \frac{x-y}{(x+y)^3}dx\right\}dy.$$

Solution. L.H.S. $=\int_0^1\left\{\int_0^1 \frac{x-y}{(x+y)^3}dy\right\}dx=\frac{1}{2}$, by above example 3.

For R.H.S., $\frac{x-y}{(x+y)^3}=\frac{(x+y-y)-y}{(x+y)^3}=\frac{(x+y)-2y}{(x+y)^3}=\frac{1}{(x+y)^2}-\frac{2y}{(x+y)^3}$, so we have

$$\text{R.H.S.}=\int_0^1\left\{\int_0^1\left(\frac{1}{(x+y)^2}-\frac{2y}{(x+y)^3}\right)dx\right\}dy=\int_0^1\left[-\frac{1}{(x+y)}+\frac{y}{(x+y)^2}\right]_0^1 dy$$

$$=\int_0^1\left[\left\{-\frac{1}{(1+y)}+\frac{y}{(1+y)^2}\right\}-\left\{-\frac{1}{y}+\frac{1}{y}\right\}\right]dy=-\int_0^1\frac{dx}{(1+y)^2}=\left[\frac{1}{1+y}\right]_0^1=\left[\frac{1}{1+1}-\frac{1}{0+1}\right]=-\frac{1}{2}.$$

$\Rightarrow$ L.H.S. $\neq$ R.H.S.

Hence $\int_0^1\left\{\int_0^1 \frac{x-y}{(x+y)^3}dy\right\}dx\neq\int_0^1\left\{\int_0^1 \frac{x-y}{(x+y)^3}dx\right\}dy$. Proved.

Example 5. Evaluate $\int_0^1\int_0^{\sqrt{1+x^2}}\frac{1}{(1+x^2+y^2)}dydx$.

Solution. Clearly, here $y = f(x)$ varies from 0 to $\sqrt{1+x^2}$ and finally x (as an independent variable) goes from 0 to 1. So we have

$$\int_0^1\int_0^{\sqrt{1+x^2}} \frac{1}{(1+x^2+y^2)}\,dy\,dx = \int_0^1\left(\int_0^{\sqrt{1+x^2}} \frac{1}{(1+x^2)+y^2}\,dy\right)dx$$

$$= \int_0^1\left(\int_0^{\sqrt{1+x^2}} \frac{1}{a^2+y^2}\,dy\right)dx,\ a^2 = (1+x^2)$$

$$= \int_0^1\left(\frac{1}{a}\tan^{-1}\frac{y}{a}\right)_0^{\sqrt{1+x^2}} dx$$

$$= \int_0^1 \frac{1}{\sqrt{1+x^2}}\left(\tan^{-1}\frac{\sqrt{1+x^2}}{\sqrt{1+x^2}} - \tan^{-1}0\right)dx$$

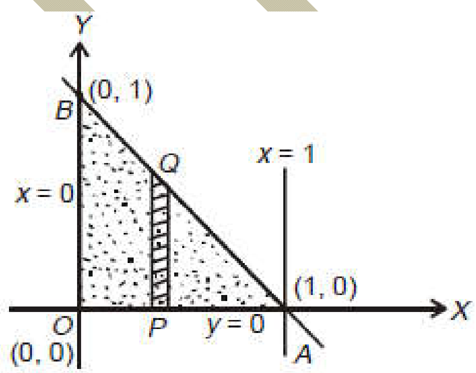
$$= \int_0^1 \frac{1}{\sqrt{1+x^2}}\left(\frac{\pi}{4} - 0\right)dx = \frac{\pi}{4}\left[\log\left\{x+\sqrt{1+x^2}\right\}\right]_0^1$$

$$= \frac{\pi}{4}\log(1+\sqrt{2})$$

Ans.

Example 6. Evaluate $\iint_R e^{2x+3y}\,dxdy$ over the triangle bounded by the lines $x = 0$, $y = 0$ and $x + y = 1$.

Solution. Here the region of integration is the triangle $OABO$ as the line $x + y = 1$ intersects the axes at points (1, 0) and (0, 1). Thus, precisely the region R (say) can be expressed as: $x = 0$ to 1 and $y = 0$ to $1 - x$ as in the figure given below:



$$\therefore\quad \iint_R e^{2x+3y}\,dxdy = \int_0^1\left(\int_0^{1-x} e^{2x+3y}\,dy\right)dx = \int_0^1\left[\frac{1}{3}e^{2x+3y}\right]_0^{1-x} dx$$

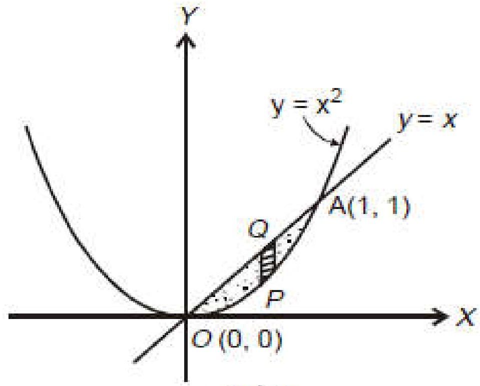
$$= \frac{1}{3}\int_0^1 (e^{3-x} - e^{2x})\,dx = \frac{1}{3}\left[\frac{e^{3-x}}{-1} - \frac{e^{2x}}{2}\right]_0^1$$

$$= \frac{-1}{3}\left[\left(e^2 + \frac{e^2}{2}\right) - \left(e^3 + \frac{1}{2}\right)\right] = \frac{1}{6}\left[2e^3 - 3e^2 + 1\right]$$

$$= \frac{1}{6}\left[(2e+1)(e-1)^2\right].$$ Ans.

Example 7. Evaluate the integral $\iint_R xy(x+y)\,dxdy$ over the area between the curves $y = x^2$ and $y = x$.

Solution. We have $y = x^2$ and $y = x$ which implies $x^2 - x = 0$ i.e. either $x = 0$ or $x = 1$. Further, if $x = 0$ then $y = 0$; if $x = 1$ then $y = 1$. Means the two curves intersect at points (0, 0), (1, 1). The region R of integration is doted and can be expressed as: $x = 0$ to 1 and $y = x^2$ to x in the figure given below:



$$\therefore \quad \iint_R xy(x+y)\,dxdy = \int_0^1\left(\int_{x^2}^{x} xy(x+y)\,dy\right)dx$$

$$= \int_0^1\left\{\left(x^2\frac{y^2}{2} + x\frac{y^3}{3}\right)_{x^2}^{x}\right\}dx$$

$$= \int_0^1\left\{\left(\frac{x^4}{2}+\frac{x^4}{3}\right)-\left(\frac{x^6}{2}+\frac{x^7}{3}\right)\right\}dx$$

$$= \int_0^1\left(\frac{5}{6}x^4 - \frac{1}{2}x^6 - \frac{1}{3}x^7\right)dx$$

$$= \left[\frac{5}{6}\times\frac{x^5}{5} - \frac{1}{2}\frac{x^7}{7} - \frac{1}{3}\frac{x^8}{8}\right]_0^1 = \frac{1}{6} - \frac{1}{14} - \frac{1}{24} = \frac{3}{56}.$$ Ans.

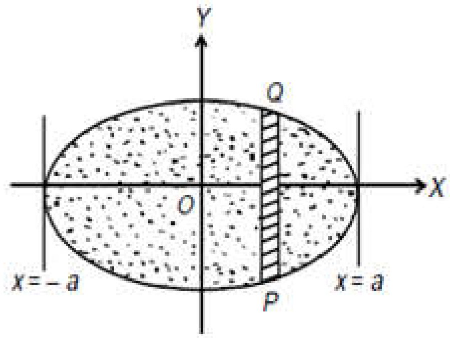
Example 8. Evaluate $\iint (x+y)^2\,dxdy$ over the area bounded by the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$.

Solution. For the given ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$, the region of integration can be considered as bounded by the curves $y = -b\sqrt{1-\frac{x^2}{a^2}}$, $y = b\sqrt{1-\frac{x^2}{a^2}}$ and finally x goes from $-a$ to a

$$\therefore \quad I = \iint (x+y)^2\,dxdy = \int_{-a}^{a}\left(\int_{-b\sqrt{1-x^2/a^2}}^{b\sqrt{1-x^2/a^2}}(x^2+y^2+2xy)\,dy\right)dx = \int_{-a}^{a}\left(\int_{-b\sqrt{1-x^2/a^2}}^{b\sqrt{1-x^2/a^2}}(x^2+y^2)\,dy\right)dx$$

Here $\int 2xy\,dy = 0$ as it has the same integral value for both limits i.e., the term xy, which is an odd function of y, on integration gives a zero value. So we have

$$I = 4\int_0^a \left(\int_0^{b\sqrt{1-x^2/a^2}} \left(x^2+y^2\right) dy \right) dx$$



$$I = 4\int_0^a \left[x^2 y + \frac{y^3}{3} \right]_0^{b\sqrt{1-x^2/a^2}} dx$$

$$\Rightarrow I = 4\int_0^a \left[x^2 b\left(1-\frac{x^2}{a^2}\right)^{1/2} + \frac{b^3}{3}\left(1-\frac{x^2}{a^2}\right)^{3/2} \right] dx$$

On putting $x = a\sin\theta$, $dx = a\cos\theta\, d\theta$; we get

$$I = 4b\int_0^{\pi/2} \left(\left(a^2\sin^2\theta\cos\theta\right) + \frac{b^3}{3}\cos^3\theta \right) a\cos\theta\, d\theta$$

$$= 4ab\int_0^{\pi/2} \left(a^2\sin^2\theta\cos^2\theta + \frac{b^3}{3}\cos^4\theta \right) d\theta$$

Now using formula $\displaystyle\int_0^{\pi/2} \sin^p x\cos^q x\, dx = \frac{\frac{1}{2}\Gamma\left(\frac{p+1}{2}\right)\Gamma\left(\frac{q+1}{2}\right)}{\Gamma\left(\frac{p+q+2}{2}\right)}$

and $\displaystyle\int_0^{\pi/2} \cos^n x\, dx = \frac{\Gamma\left(\frac{n+1}{2}\right)}{\Gamma\left(\frac{n+2}{2}\right)}\frac{\sqrt{\pi}}{2}$, (in particular when $p=0$, $q=n$)

$$\iint (x+y)^2\, dx\, dy = 4ab\left\{ a^2 \frac{\Gamma\frac{3}{2}\,\Gamma\frac{3}{2}}{2\Gamma 3} + \frac{b^2}{3}\,\frac{\Gamma\frac{5}{2}\,\Gamma\frac{1}{2}}{2\Gamma 3} \right\}$$

$$= 4ab\left\{ a^2 \frac{\frac{\sqrt{\pi}}{2}\,\frac{\sqrt{\pi}}{2}}{2.2.1} + \frac{b^2}{3}\,\frac{\frac{3}{2}\,\frac{\sqrt{\pi}}{2}\sqrt{\pi}}{2.2.1} \right\} \qquad \left| \because \Gamma\left(\frac{1}{2}\right) = \sqrt{\pi} \right.$$

$$= 4ab\left\{ \frac{\pi a^2}{16} + \frac{\pi b^2}{16} \right\} = \frac{\pi ab\left(a^2+b^2\right)}{4}$$

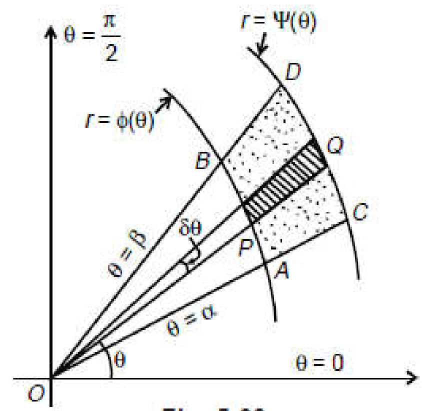
. Ans.

Double Integrals (Polar)

To evaluate $\int_{\theta=\alpha}^{\theta=\beta}\int_{r=\phi(\theta)}^{r=\Psi(\theta)} f(r,\theta)\ dr\ d\theta$, we first integrate with respect to r between the limits $r=\phi(\theta)$ to $r=\psi(\theta)$ keeping θ as a constant and then the resulting expression is integrated with respect to θ from $\theta=\alpha$ to $\theta=\beta$.

Geometrical Illustration: Let AB and CD be the two continuous curves $r=\phi(\theta)$ and $r=\Psi(\theta)$ bounded between the lines $\theta=\alpha$ and $\theta=\beta$ so that $ABDC$ is the required region of integration.

Let PQ be a radial strip of angular thickness $\delta\theta$ when OP makes an angle θ with the initial line.

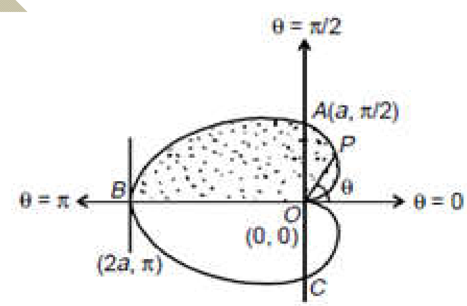


Here $\int_{r=\phi(\theta)}^{r=\Psi(\theta)} f(r,\theta)dr$ refers to the integration with

Example 9. Evaluate $\iint r\sin\theta\, dr\, d\theta$ over the cardiod $r=a(1-\cos\theta)$ above the initial line.

Solution. The region of integration under consideration is the cardiod $r=a(1-\cos\theta)$ above the initial line. In the cardiod $r=a(1-\cos\theta)$;

$$\left.\begin{array}{lll} \text{for} & \theta=0, & r=0, \\ \text{for} & \theta=\dfrac{\pi}{2}, & r=a, \\ \text{for} & \theta=\pi, & r=2a \end{array}\right\}$$



As clear from the geometry along the radial strip OP, r (as a function of θ) varies from $r=0$ to $r=a(1-\cos\theta)$ and then this strip slides from $\theta=0$ to $\theta=\pi$ for covering the area above the initial line.

Hence

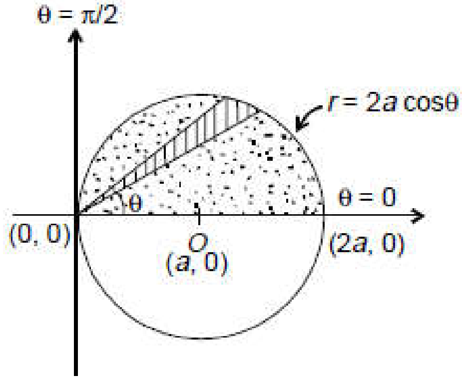
$$\iint r\sin\theta\, dr\, d\theta = \int_0^{\pi}\left(\int_0^{r=a(1-\cos\theta)} r\, dr\right)\sin\theta\, d\theta = \int_0^{\pi}\left(\frac{r^2}{2}\Bigg|_0^{a(1-\cos\theta)}\right)\sin\theta\, d\theta$$

$$= \frac{a^2}{2}\int_0^{\pi}(1-\cos\theta)^2\sin\theta\, d\theta = \frac{a^2}{2}\left[\frac{(1-\cos\theta)^3}{3}\right]_0^{\pi}, \quad \because \int f^n(x)\,f'(x)\,dx = \frac{f^{n+1}(x)}{n+1}$$

$$= \frac{a^2}{6}\left[(1-\cos\pi)^3-(1-\cos 0)\right] = \frac{a^2}{6}[8-0] = \frac{4a^2}{3}. \quad \text{Ans.}$$

Example 10. Show that $\iint_R r^2\sin\theta\, dr\, d\theta = \frac{2a^3}{3}$, where R is the semi circle $r=2a\cos\theta$ above the initial line.

Solution. The region R of integration is the semi-circle $r=2a\cos\theta$ above the initial line.



For the circle $\left.\begin{array}{l} r = 2a\cos\theta,\ \theta = 0 \quad \Rightarrow \quad r = 2a \\ \theta = \dfrac{\pi}{2} \quad \Rightarrow \quad r = 0 \end{array}\right\}$

Otherwise also, $r = 2a\cos\theta \quad \Rightarrow \quad r^2 = 2ar\cos\theta$

$$x^2 + y^2 = 2ax$$

$$(x^2 - 2ax + a^2) + y^2 = a^2$$

$$(x - a)^2 + (y - 0)^2 = a^2$$

Hence the required integral is

$$\int_0^{\frac{\pi}{2}}\int_0^{2a\cos\theta} r^2 \sin\theta\, dr\, d\theta = \int_0^{\frac{\pi}{2}}\left(\int_0^{2a\cos\theta} r^2 dr\right)\sin\theta\, d\theta = \int_0^{\pi/2}\left(\left|\frac{r^3}{3}\right|_0^{2a\cos\theta}\right)\sin\theta\, d\theta$$

$$= \frac{-1}{3}\int_0^{\pi/2}(2a)^3\cos^3\theta\sin\theta\, d\theta$$

$$= \frac{-8a^3}{3}\left(\frac{\cos^4\theta}{4}\right)_0^{\pi/2}, \quad \text{using } \int f^n(x)\ f'(x)dx = \frac{f^{n+1}(x)}{n+1}$$

$$= \frac{2a^3}{3}. \quad \underline{\text{Proved.}}$$

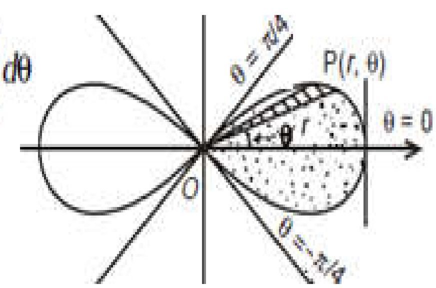
Example 11. Evaluate $\displaystyle\iint \frac{r\, dr\, d\theta}{\sqrt{a^2 + r^2}}$ over one loop of the lemniscate $r^2 = a^2\cos 2\theta$.

Solution. The lemniscate is bounded. For $r = 0, \theta = \pm\dfrac{\pi}{4}$ and maximum value of r is a. See below figure, in one complete loop, r varies from 0 to $r = a\sqrt{\cos 2\theta}$ and the radial strip slides between $\theta = -\dfrac{\pi}{4}$ to $\dfrac{\pi}{4}$.

Hence the required integral is

$$\int_{-\pi/4}^{\pi/4}\int_0^{a\sqrt{\cos 2\theta}} \frac{r}{(a^2 + r^2)^{1/2}}\, dr\, d\theta = \int_{-\pi/4}^{\pi/4}\left(\int_0^{a\sqrt{\cos 2\theta}} d(a^2 + r^2)^{1/2}\, dr\right) d\theta$$

$$= \int_{-\pi/4}^{\pi/4} \left.(a^2 + r^2)^{1/2}\right|_0^{a\sqrt{\cos 2\theta}} d\theta$$



$$= \int_{-\pi/4}^{\pi/4}\left[\left(a^2 + a^2\cos 2\theta\right)^{1/2} - a\right]d\theta = a\int_{-\pi/4}^{\pi/4}\left(\sqrt{2}\cos\theta - 1\right)d\theta$$

$$= 2a\int_0^{\pi/4}\left(\sqrt{2}\cos\theta - 1\right)d\theta = 2a\left[\left(\sqrt{2}\sin\theta - \theta\right)_0^{\pi/4}\right]$$

$$= 2a\left[\sqrt{2}\frac{1}{\sqrt{2}} - \frac{\pi}{4}\right] = 2a\left(1 - \frac{\pi}{4}\right). \text{ Ans.}$$

Example 12. Evaluate $\int\int r^3 dr\, d\theta$, over the area included between the circles $r = 2a\cos\theta$ and $r = 2b\cos\theta$ ($b < a$).

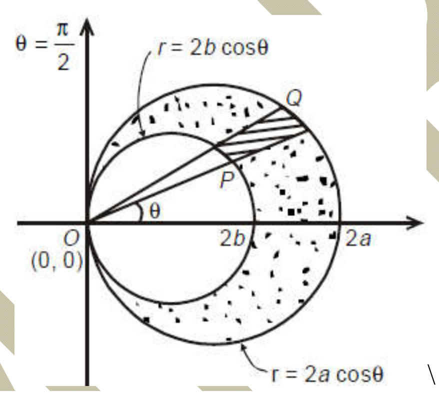
Solution. Given $r = 2a\cos\theta$ or $r^2 = 2ar\cos\theta$

or $x^2 + y^2 = 2ax$ or $(x + a)^2 + (y - 0)^2 = a^2$

i.e this curve represents the circle with centre $(a, 0)$ and radius a.

Likewise, $r = 2b\cos\theta$ represents the circle with centre $(b, 0)$ and radius b.

The area of integration is bounded by two circles $r = 2a\cos\theta$ and $r = 2b\cos\theta$ ($b < a$), where over the radical strip PQ, r varies from circle $r = 2b\cos\theta$ to circle $r = 2a\cos\theta$ and finally θ varies from $-\pi/2$ to $\pi/2$.



Thus, the given integral $\int\int_R r^3 dr\, d\theta = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \int_{2b\cos\theta}^{2a\cos\theta} r^3 dr\, d\theta$

$$= \int_{-\pi/2}^{\pi/2}\left[\frac{r^4}{4}\right]_{2b\cos\theta}^{2a\cos\theta} d\theta$$

$$= \frac{1}{4}\int_{-\pi/2}^{\pi/2}\left[(2a\cos\theta)^4 - (2b\cos\theta)^4\right]d\theta$$

$$= 4(a^4 - b^4)\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos^4\theta\, d\theta$$

$$= 8(a^4 - b^4)\int_0^{\frac{\pi}{2}} \cos^4\theta\, d\theta$$

$$= 8(a^4 - b^4)\frac{3\cdot 1}{4\cdot 2}\frac{\pi}{2}$$

$$= \frac{3}{2}\pi(a^4 - b^4).$$

Particular Case: When $r = 2\cos\theta$ and $r = 4\cos\theta$ *i.e.*, $a = 2$ and $b = 1$, then

$$I = \frac{3}{2}\pi(a^4 - b^4) = \frac{3}{2}\pi(2^4 - 1^4) = \frac{45\pi}{2} \text{ units.}$$

LECTURE NO.:- 28

Double Integrals by Changing into Polar Form

Sometimes the evaluation of double integral becomes more convenient by a suitable change of variables from one system to another system.

The cartesian variables x and y may be changed to polar variables r and θ by the relations $x = r\cos\theta$ and $y = r\sin\theta$.

Let us have double integral $I = \iint_R f(x, y)dxdy$.

To change the variables from cartesian to polar form, putting $x = r\cos\theta$ and $y = r\sin\theta$, the double integral is transformed to

$$I = \iint_R f(r\cos\theta, r\sin\theta)Jdrd\theta,$$

where $J = \dfrac{\partial(x,y)}{\partial(r,\theta)} = \begin{vmatrix} \dfrac{\partial x}{\partial r} & \dfrac{\partial x}{\partial \theta} \\ \dfrac{\partial y}{\partial r} & \dfrac{\partial y}{\partial \theta} \end{vmatrix} = \begin{vmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{vmatrix} = r$

Thus we get $I = \iint_R f(x, y)dxdy = \iint_R f(r\cos\theta, r\sin\theta)rdrd\theta$.

Note: This change is especially useful when the region of integration is a circle or a part of circle.

Example 1. Transform the integral $\int_0^\infty \int_0^\infty e^{-(x^2+y^2)}dxdy$ to polar coordinates and hence evaluate it. Also show that $\int\limits_0^\infty e^{-x^2}dx = \dfrac{\sqrt{\pi}}{2}$.

Solution. Let $I = \int_0^\infty \int_0^\infty e^{-(x^2+y^2)}dxdy$

Here the limits of x and y are both from 0 to ∞.

So we are to consider only the first quadrant.

The cartesian variables are changed to polar form by putting $x = r\cos\theta$ and $y = r\sin\theta$. In the first quadrant r varies from 0 to ∞ and θ varies from 0 to π/2.

Hence the double integral is transformed to

$$I=\int_0^\infty\int_0^\infty e^{-(x^2+y^2)}dxdy=\int_0^\infty\int_0^{\frac{\pi}{2}} e^{-r^2}rdrd\theta$$

$$=[\theta]_0^{\pi/2}\left[\frac{e^{-r^2}}{2}\right]_0^\infty=\frac{\pi}{4}.$$ Ans.

Again

$$I=\int_0^\infty\int_0^\infty e^{-(x^2+y^2)}dxdy=\int_0^\infty e^{-x^2}dx\int_0^\infty e^{-y^2}dy$$

$$=\left[\int_0^\infty e^{-x^2}dx\right]^2=\frac{\pi}{4}$$

Hence $\int\limits_0^\infty e^{-x^2}dx=\frac{\sqrt{\pi}}{2}$. Ans.

Example 2. Evaluate $\int_0^1\int_x^{\sqrt{2x-x^2}}\sqrt{x^2+y^2}\,dxdy$ by changing to polar coordinates.

Solution. Here $x\le y\le\sqrt{2x-x^2}$ & $0\le x\le 1$.

The region of integration is bounded by the line $y=x$ and the circle $y^2=2x-x^2$, i.e. $x^2-2x+y^2=0$, i.e. $(x-1)^2+y^2=1$.

Changing to polar coordinates, we put $x=r\cos\theta$ and $y=r\sin\theta$.

Then we have

$$x=0\Rightarrow r\cos\theta=0\Rightarrow r=0,\theta=\frac{\pi}{2};$$

$$y^2=2x-x^2\Rightarrow x^2+y^2=2x\Rightarrow r^2=2r\cos\theta\Rightarrow r=2\cos\theta;$$

$$y=x\Rightarrow r\cos\theta=r\sin\theta\Rightarrow\theta=\frac{\pi}{4}.$$

$$\therefore I=\int_0^1\int_x^{\sqrt{2x-x^2}}\sqrt{x^2+y^2}\,dxdy$$

$$=\int_{\pi/4}^{\pi/2}\int_0^{2\cos\theta}r^2drd\theta$$

$$=\int_{\pi/4}^{\pi/2}\left[\frac{r^3}{3}\right]_0^{2\cos\theta} d\theta \quad =\frac{8}{3}\int_{\pi/4}^{\pi/2}\cos^3\theta d\theta$$

Let $\sin\theta = t \Rightarrow \cos\theta d\theta = dt$ and limit $t=\frac{1}{\sqrt{2}}$ to $t=1$.

$$\therefore I=\frac{8}{3}\int_{1/\sqrt{2}}^{1}(1-t^2)dt=\frac{8}{3}\left(t-\frac{t^3}{3}\right)_{1/\sqrt{2}}^{1}$$

$$=\frac{8}{3}\left\{1-\frac{1}{3}-\frac{1}{\sqrt{2}}+\frac{1}{2\sqrt{2}.3}\right\}=\frac{2}{9}\left(8-5\sqrt{2}\right).$$ Ans.

Example 3. Evaluate $\int_0^a\int_0^{\sqrt{a^2-y^2}}(x^2+y^2)dxdy$ by changing to polar coordinates.

Solution. Let $I=\int_0^a\int_0^{\sqrt{a^2-y^2}}(x^2+y^2)dxdy$

Changing cartesian to polar, we have

$$I=\int_{\theta=0}^{\pi/2}\int_{r=0}^{a}r^3drd\theta \;=\int_{\theta=0}^{\pi/2}\left[\frac{r^4}{4}\right]_0^a d\theta =\frac{a^4}{4}[\theta]_0^{\pi/2}=\frac{\pi a^4}{8}.$$ Ans.

Example 4. Evaluate $\int_0^a\int_x^{\sqrt{2ax-x^2}}\left(x^2+y^2\right)dxdy$ by changing to polar coordinates.

Solution. Try yourself.

Example 5. Evaluate $\int_0^a\int_y^a\frac{xdxdy}{\left(x^2+y^2\right)}$ by changing to polar coordinates.

Solution. Try yourself.

Example 6. Transform the integral $\int_0^a\int_0^{\sqrt{a^2-x^2}}y^2\sqrt{(x^2+y^2)}dxdy$ by changing to polar coordinates and hence evaluate it.

Solution. Try yourself.

Example 7. Change $\int_0^{4a}\int_{y^2/4a}^{y}\frac{\left(x^2-y^2\right)}{\left(x^2+y^2\right)}dxdy$ to polar coordinates.

Solution. Try yourself.

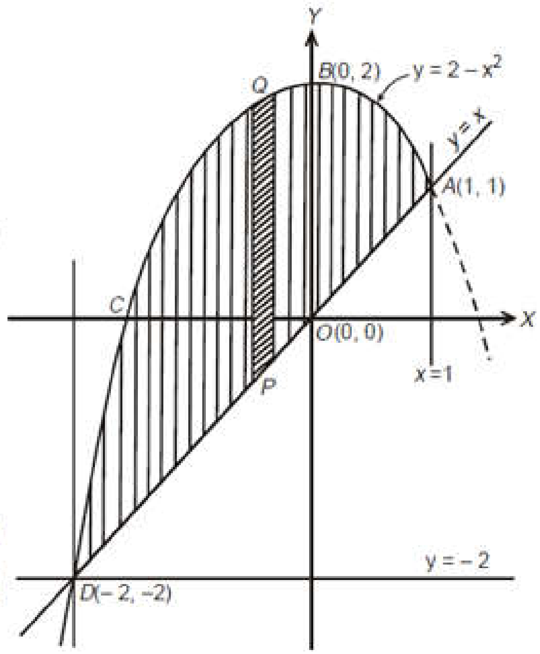
LECTURE NO.:- 29

Areas by Double Integration

Example 1. Find by double integration, the area lying between the curves $y = 2 - x^2$ and $y = x$.

Solution. The given curve $y = 2 - x^2$ is a parabola with vertex at the point (0, 2) and curve $y = x$ is a straight line. Parabola may be written as $x^2 = -(y - 2)$,

where in $\left.\begin{array}{rcr} x=0 & \Rightarrow & y=2 \\ x=1 & \Rightarrow & y=1 \\ x=2 & \Rightarrow & y=-2 \\ x=-1 & \Rightarrow & y=1 \\ x=-2 & \Rightarrow & y=-2 \end{array}\right\}$



i.e., it passes through points (0, 2), (1, 1), (2, − 2), (− 1, 1), (− 2, − 2).

Likewise, the curve $y = x$ is a straight line

where $\left.\begin{array}{rcr} y=0 & \Rightarrow & x=0 \\ y=1 & \Rightarrow & x=1 \\ y=-2 & \Rightarrow & x=-2 \end{array}\right\}$

i.e., it passes through (0, 0), (1, 1), (− 2, − 2)

Now for the two curves $y = x$ and $y = 2 - x^2$ to intersect, $x = 2 - x^2$ or $x^2 + x - 2 = 0$ i.e., $x = 1, -2$ which in turn implies $y = 1, -2$ respectively.

Thus, the two curves intersect at (1, 1) and (−2, −2).

Clearly, the area need to be required is $ABCDA$.

$$\therefore \qquad A = \int_{-2}^{1}\left(\int_{x}^{2-x^2} dy\right)dx = \int_{-2}^{1}(2 - x^2 - x)\,dx$$

$$= \left[2x - \frac{x^3}{3} - \frac{x^2}{2}\right]_{-2}^{1} = \frac{9}{2} \text{ units.}$$

<u>Ans.</u>

Example 2. Find by double integration, the area lying between the parabola $y = 4x - x^2$ and the line $y = x$.

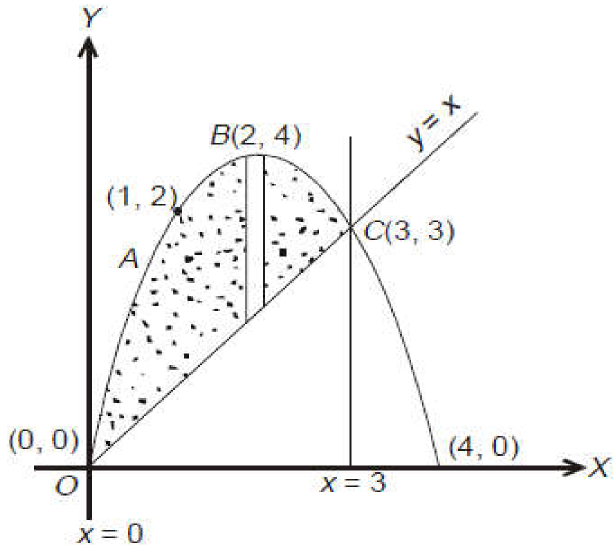
Solution. For the given curve $y = 4x - x^2$, we have

$\left.\begin{array}{rcr} x=0 & \Rightarrow & y=0 \\ x=1 & \Rightarrow & y=2 \\ x=2 & \Rightarrow & y=4 \\ x=3 & \Rightarrow & y=3 \\ x=4 & \Rightarrow & y=0 \end{array}\right\}$

i.e. it passes through the points (0, 0), (1, 2), (3, 3) and (4, 0).

Likewise, the curve $y = x$ passes through (0, 0) and (3, 3), and hence, (0, 0) and (3, 3) are the common points.

Otherwise also putting $y = x$ into $y = 4x - x^2$, we get $x = 4x - x^2 \Rightarrow x = 0, 3$.



See in above figure, *OABCO* is the area bounded by the two curves $y = x$ and $y = 4x - x^2$

$$\therefore \text{ Area } \quad OABCO = \int_0^3 \int_x^{4x-x^2} dy\,dx$$

$$= \int_0^3 [y]_x^{4x-x^2}\,dx$$

$$= \int_0^3 (4x - x^2 - x)\,dx = \left[3\frac{x^2}{2} - \frac{x^3}{3}\right]_0^3$$

$= 9/2$ units. <u>Ans.</u>

Example 3. Calculate the area of the region bounded by the curves

$y = 3x/(x^2+2)$ and $4y = x^2$.

Solution. The curve $4y = x^2$ is a parabola,

where $\left.\begin{array}{l} y = 0 \Rightarrow x = 0, \\ y = 1 \Rightarrow x = \pm 2 \end{array}\right\}$ *i.e.*, it passes through (–2, 1), (0, 0), (2, 1).

Likewise, for the curve $y = \dfrac{3x}{x^2+2}$

$$\left.\begin{array}{lll} y = 0 & \Rightarrow & x = 0 \\ y = 1 & \Rightarrow & x = 1, 2 \\ x = -1 & \Rightarrow & y = -1 \end{array}\right\}$$

Hence it passes through points (0, 0), (1, 1), (2, 1), (–1, –1).

Also for the curve $(x^2 + 2)\,y = 3x$, $y = 0$ (i.e. X-axis) is an asymptote.

For the points of intersection of the two curves $y = \dfrac{3x}{x^2+2}$ and $4y = x^2$

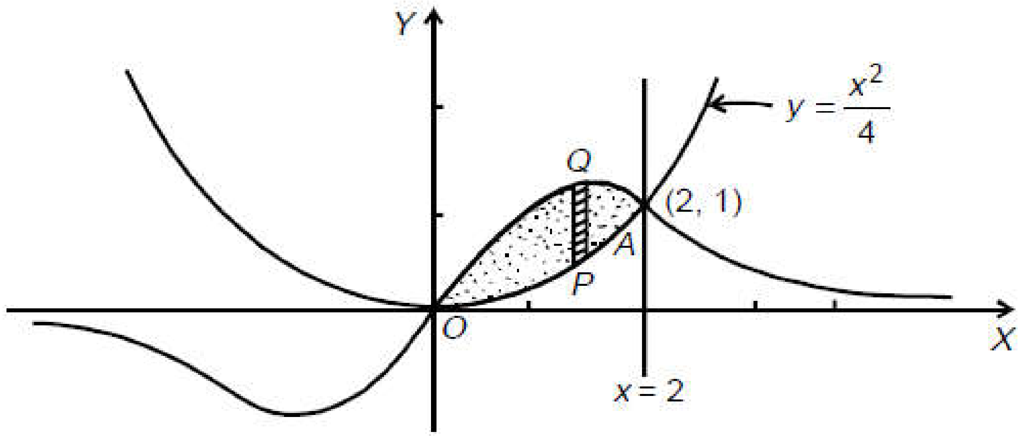
we write $\dfrac{3x}{x^2+2} = \dfrac{x^2}{4}$ or $x^2(x^2+2) = 12x$

$\Rightarrow x(x-2)(x^2+2x+6) = 0 \Rightarrow x = 0, x = 2.$

Then $x = 0 \Rightarrow y = 0$

$x = 2 \Rightarrow y = 1$

i.e. (0, 0) and (2, 1) are the two points of intersection.



Hence the required area is

The area under consideration,

$$A = \int_0^2 \left(\int_{y=\frac{x^2}{4}}^{y=\frac{3x}{x^2+2}} dy \right) dx = \int_0^2 \left[\frac{3x}{x^2+2} - \frac{x^2}{4} \right] dx$$

$$= \left[\frac{3}{2}\log(x^2+2) - \frac{x^3}{12} \right]_0^2$$

$$= \frac{3}{2}(\log 6 - \log 2) - \frac{2}{3} = \log 3^{\frac{3}{2}} - \frac{2}{3}.$$

Ans.

Example 4. Find by the double integration, the area lying inside the circle $r = a \sin\theta$ and outside the cardiod $r = a(1 - \cos\theta)$.

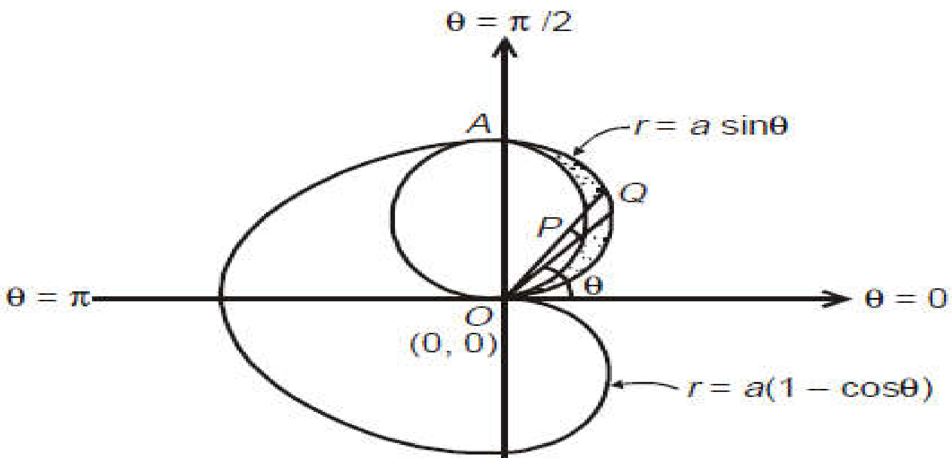
Solution. For the circle $r = a \sin\theta$, we have

$$\left.\begin{array}{l} \theta = 0 \implies r = 0 \\ \theta = \frac{\pi}{2} \implies r = a \\ \theta = \pi \implies r = 0 \end{array}\right\}$$

Likewise for the cardiod $r = a(1 - \cos\theta)$:

$$\left.\begin{array}{l} \theta = 0 \implies r = 0 \\ \theta = \frac{\pi}{2} \implies r = a \\ \theta = \pi \implies r = 2a \end{array}\right\}$$

The area enclosed inside the circle $r = a\sin\theta$ and outside the cardiod $r = a(1 - \cos\theta)$ is shown as doted one in figure. For the radial strip PQ, r varies from $r = a(1 - \cos\theta)$ to $r = a\sin\theta$ and finally θ varies in between 0 to $\pi/2$.



Thus the two curves intersect at θ = 0 and θ = $\frac{\pi}{2}$.

$$A=\int_0^{\frac{\pi}{2}}\int_{a(1-\cos\theta)}^{a\sin\theta} r\,dr\,d\theta$$

$$=\int_0^{\pi/2}\frac{r^2}{2}\Bigg|_{a(1-\cos\theta)}^{a\sin\theta}d\theta$$

$$=\int_0^{\pi/2}\frac{1}{2}\left[\sin^2\theta-\left(1+\cos^2\theta-2\cos\theta\right)\right]d\theta$$

$$=\frac{a^2}{2}\int_0^{\pi/2}[-\cos 2\theta-1+2\cos\theta]d\theta,\ \text{ since }\ (\sin^2\theta-\cos^2\theta)=-\cos 2\theta$$

$$=\frac{a^2}{2}\left[\frac{-\sin 2\theta}{2}-\theta+2\sin\theta\right]_0^{\pi/2}=a^2\left(1-\frac{\pi}{4}\right).$$ Ans.

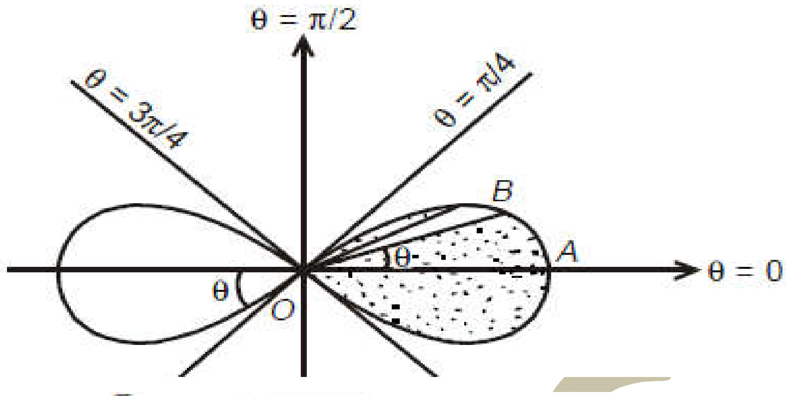
Example 5. Find by double integration, the area of laminiscate $r^2 = a^2 \cos 2\theta$.

Solution. As the given curve $r^2 = a^2 \cos 2\theta$ contains cosine terms only and hence it is symmetrical about the initial axis.

Further the curve lies wholly inside the circle $r = a$, since the maximum value of $|\cos\theta|$ is 1.

Also, no portion of the curve lies between $\theta=\frac{\pi}{4}$ to $\theta=\frac{3\pi}{4}$ and the extended axis.

See the geometry, for one loop, the curve is bounded between $\theta=-\frac{\pi}{4}$ to $\frac{\pi}{4}$



$$\therefore \quad \text{Area} = 2\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \int_{r=0}^{r=\sqrt{a^2\cos 2\theta}} r\,dr\,d\theta$$

$$= 4\int_0^{\pi/4} \frac{r^2}{2}\bigg|_0^{a\sqrt{\cos 2\theta}} d\theta$$

$$= 2a^2\int_0^{\pi/4} \cos 2\theta\, d\theta = 2a^2\left[\frac{\sin 2\theta}{2}\right]_0^{\pi/4} = a^2. \quad \underline{\text{Ans.}}$$

Example 6. Calculate the area included between the curve $r = a(\sec\theta + \cos\theta)$ and its asymptote $r = a\sec\theta$.

Solution. The equation of given curve $r = a(\sec\theta + \cos\theta)$ is uneffected by replacing θ by −θ. So it is symmetrical about initial line (x-axis).

Further on initial line $\theta = 0$, $r = 2a$ and when $\theta \to \pi/2$, $r \to \infty$.

Clearly, the required area is the doted region in which r varies along the radial strip from $r = a\sec\theta$ to $r = a(\sec\theta + \cos\theta)$ and finally strip slides between $\theta = -\frac{\pi}{2}$ to $\theta = \frac{\pi}{2}$.

$$\therefore \quad A = 2\int_0^{\frac{\pi}{2}} \int_{a\sec\theta}^{a(\sec\theta+\cos\theta)} r\,dr\,d\theta$$

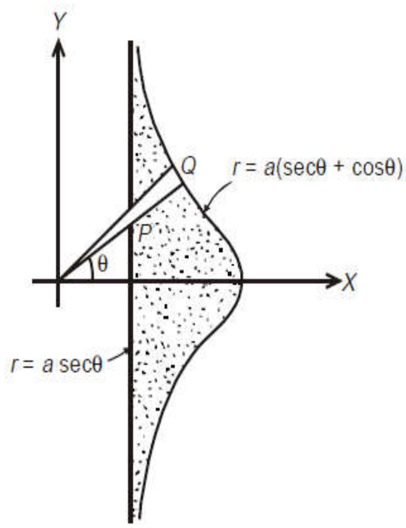
$$= 2\int_0^{\pi/2} \left[\frac{r^2}{2}\right]_{a\sec\theta}^{a(\sec\theta+\cos\theta)} d\theta$$

$$= a^2\int_0^{\pi/2} \left[\left(\frac{1+\cos^2\theta}{\cos\theta}\right)^2 - \left(\frac{1}{\cos\theta}\right)^2\right] d\theta$$

$$= a^2\int_0^{\pi/2} (\cos^2\theta + 2)\,d\theta$$

$$= a^2\int_0^{\pi/2} \frac{(5+\cos 2\theta)}{2}\,d\theta$$

$$= \frac{a^2}{2}\left[5\theta + \frac{\sin 2\theta}{2}\right]_0^{\pi/2} = \frac{5\pi a^2}{4}. \quad \underline{\text{Ans.}}$$



LECTURE NO.:- 30

Volumes by Double Integration

Volume V = $\iint_R f(x, y)\,dx\,dy$,

where R is the region of integration in the xy -plane under the surface $z = f(x, y)$.

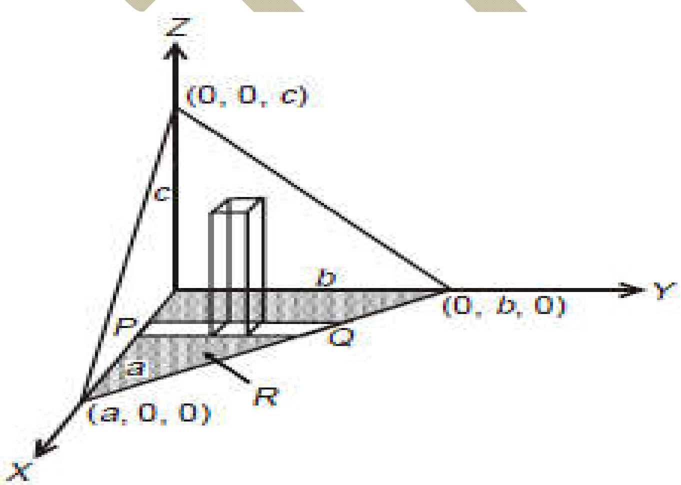
Example 7. Find the volume of the tetrahedron bounded by the plane $\frac{x}{a}+\frac{y}{b}+\frac{z}{c}=1$ and the co-ordinate planes.

Solution. Let V be the volume of the tetrahedron bounded by the plane $\frac{x}{a}+\frac{y}{b}+\frac{z}{c}=1$ and the coordinate planes $x = 0$, $y = 0$ and $z = 0$.

$\frac{x}{a}+\frac{y}{b}+\frac{z}{c}=1 \Rightarrow z = c\left(1-\frac{x}{a}-\frac{y}{b}\right) = f(x, y)$ (say).

Here $f(x, y)$ is a continuous and single valued function over the region R (say) in the xy-plane.

The intersection of the plane $\frac{x}{a}+\frac{y}{b}+\frac{z}{c}=1$ and plane $z = 0$ (xy-plane) is the line $\frac{x}{a}+\frac{y}{b}=1$ in the xy-plane. A strip PQ may be taken in the triangular base as shown in the figure. So for this strip, y varies from 0 to $b\left(1-\frac{x}{a}\right)$. For whole region R, such strips are taken from $x = 0$ to a.



Hence the required volume is

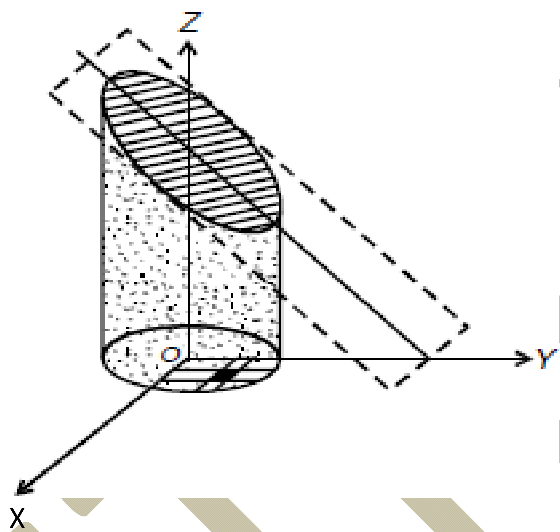
$$V = \int_0^a \int_0^{b\left(1-\frac{x}{a}\right)} c\left(1-\frac{x}{a}-\frac{y}{b}\right) dy\,dx$$

$$= c\int_0^a \left(y - \frac{xy}{a} - \frac{y^2}{2b}\right)\Bigg|_0^{b(1-x/a)} dx = c\int_0^a b\left(\frac{1}{2}-\frac{x}{a}+\frac{x^2}{2a^2}\right)dx$$

$$= cb\left[\frac{x}{2}-\frac{x^2}{2a}+\frac{x^3}{6a^2}\right]_0^a = bc\left[\frac{a}{2}-\frac{a^2}{2a}+\frac{a^3}{6a^2}\right]$$

$= \frac{abc}{6}$. Ans.

Example 8. Find the volume bounded by the cylinder $x^2 + y^2 = 4$ and the planes $y + z = 4$ and $z = 0$.
Solution. As shown in the figure, $z = 4 - y$ is to be integrated over the circle $x^2 + y^2 = 4$ in the xy-plane. To cover the shaded portion, x varies from $-\sqrt{4-y^2}$ to $\sqrt{4-y^2}$ and y varies from -2 to 2.



Hence the desired volume is

$$V = \int_{-2}^{2}\int_{-\sqrt{4-y^2}}^{\sqrt{4-y^2}} z\,dx\,dy$$

$$= 2\int_{-2}^{2}\int_{0}^{\sqrt{4-y^2}} (4-y)\,dx\,dy$$

$$= 2\int_{-2}^{2}(4-y)\left(\int_{0}^{\sqrt{4-y^2}} dx\right)dy$$

$$= 2\int_{-2}^{2}(4-y)\sqrt{4-y^2}\,dy$$

$$= 2\int_{-2}^{2}\left[4\sqrt{4-y^2} - y\sqrt{4-y^2}\right]dy$$

$$= 8\int_{-2}^{2}\sqrt{4-y^2}\,dy - 0$$

(The second term vanishes as the integrand is an odd function)

$$= 8\left[\frac{y\sqrt{4-y^2}}{2} + \frac{4}{2}\sin^{-1}\frac{y}{2}\right]_{-2}^{2} = 16\pi.$$

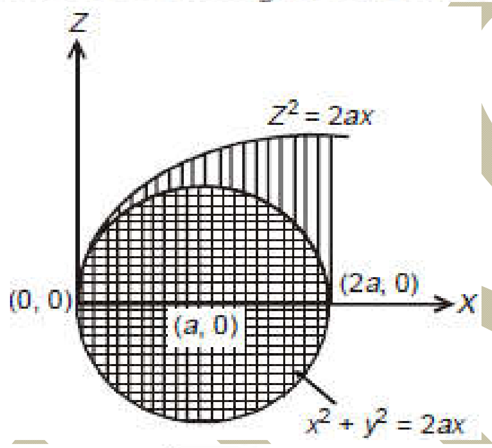
Ans.

Example 9. Prove that the volume enclosed between the cylinders $x^2 + y^2 = 2ax$ and $z^2 = 2ax$ is $\frac{128a^2}{15}$.

Solution. Let V be required volume which is enclosed by the cylinder $x^2 + y^2 = 2ax$ and the paraboloid $z^2 = 2ax$. Only half of the volume is shown in the figure.

Now, it is evident from that $z = \sqrt{2ax}$ is to be evaluated over the circle $x^2 + y^2 = 2ax$(with centre at $(a, 0)$ and radius a.

Here y varies from $-\sqrt{2ax - x^2}$ to $\sqrt{2ax - x^2}$ on the circle $x^2 + y^2 = 2ax$ and finally x varies from $x = 0$ to $x = 2a$



$$\therefore \qquad V = 2\int_0^{2a} \int_{-\sqrt{2ax-x^2}}^{\sqrt{2ax-x^2}} [z]\, dx\, dy \ \text{ as } z = f(x, y)$$

$$= 2\int_0^{2a}\left(2 \cdot \int_0^{\sqrt{2ax-x^2}} \sqrt{2ax}\right) dy\, dx$$

$$= 4\int_0^{2a} \sqrt{2ax}\left(\int_0^{\sqrt{2ax-x^2}} dy\right) dx$$

$$= 4\int_0^{2a} \sqrt{2ax}\, |y|_0^{\sqrt{2ax-x^2}}\, dx = 4\int_0^{2a} \sqrt{2ax}\sqrt{2ax - x^2}\, dx$$

$$= 4\sqrt{2a}\int_0^{2a} x\sqrt{2a - x}\, dx$$

Let $x = 2a\sin^2\theta$, so that $dx = 4a\sin\theta\cos\theta\, d\theta$. Further, for $x = 0,\ \theta = 0$; $x = 2a,\ \theta = \frac{\pi}{2}$.

$$\therefore \qquad V = 4\sqrt{2a}\int_0^{\pi/2} 2a\sin^2\theta\sqrt{2a}\cos\theta \cdot 4a\sin\theta\cos\theta\, d\theta$$

$$= 64\, a^3 \int_0^{\pi/2} \sin^3\theta\cos^2\theta\, d\theta$$

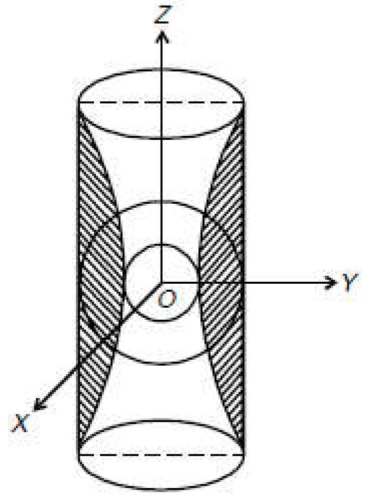
$$= 64\, a^3 \frac{(3-1)1}{5 \cdot 3} = \frac{128\, a^3}{15}.$$

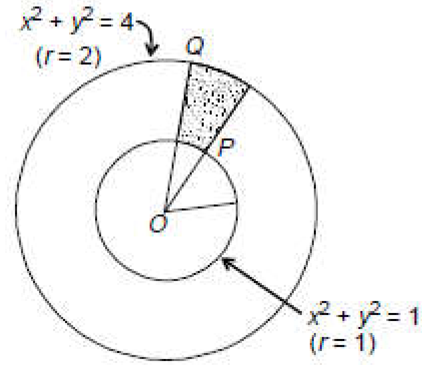
Ans.

Example 10. Find the volume bounded by the cylinder $x^2 + y^2 = 4$ and the hyperboloid $x^2 + y^2 - z^2 = 1$.

Solution. In cartesian co-ordinates, the section of the given hyperboloid $x^2 + y^2 - z^2 = 1$in the xy plane ($z = 0$) is the circle $x^2 + y^2 = 4$, whereas at the top and at the bottom end (along the z-axis *i.e.*, $z = \pm 3$) it shares common boundary with the circle $x^2 + y^2 = 4$. See figures.

Here we need to calculate the volume bounded by the two bodies (*i.e.*, the volume of shaded portion of the geometry).





(Best example of this geometry is a *solid damroo* in a *concentric long hollow drum*.)

In cylindrical polar coordinates, we see that here r varies from $r = 1$ to $r = 2$ and θ varies from 0 to 2π.

$$\therefore \quad V = 2\left[\iint z\,dx\,dy\right] = 2\left[\iint f(r,\theta)\,r\,dr\,d\theta\right]$$

$$= 2\left[\int_0^{2\pi}\int_1^2 \sqrt{r^2-1}\,r\,dr\,d\theta\right] \qquad (\because \; x^2 + y^2 - z^2 - 1 \Rightarrow z = \sqrt{x^2 + y^2 - 1})$$

$$= 2\int_0^{2\pi}\left(\int_1^2 \frac{1}{3}\,d(r^2-1)^{\frac{3}{2}}\right)d\theta$$

$$= 2\int_0^{2\pi} \left.\frac{(r^2-1)^{\frac{3}{2}}}{3}\right|_1^2 d\theta$$

$$= 2\sqrt{3}\int_0^{2\pi} d\theta = 4\pi\sqrt{3}.$$ Ans.

LECTURE NO.:- 31

Change of Order of Integration in Double Integrals (Cartesian)

Sometimes direct integration is not easy in double integrals. In such cases, if the order of integration is changed, then many double integrals are solved easily. We have seen that the double integral

$$\iint_R f(x,y)dxdy = \int_{x=a}^{x=b}\int_{y_1(x)}^{y_2(x)} f(x,y)dxdy \quad \text{.................. (1)}$$

$$= \int_{y=c}^{y=d}\int_{x_1(y)}^{x_2(y)} f(x,y)dxdy \quad \text{.................... (2)}$$

The limits of integration are determined by taking strips parallel to y -axis in the form (1) and parallel to x -axis in the form (2).

In order to change the order of integration in the double integral, first of all equations of the curves are determined from the limits of integration and then traced, from which the region of the integration is known. By geometry, the new limits of integration are determined in the desired order.

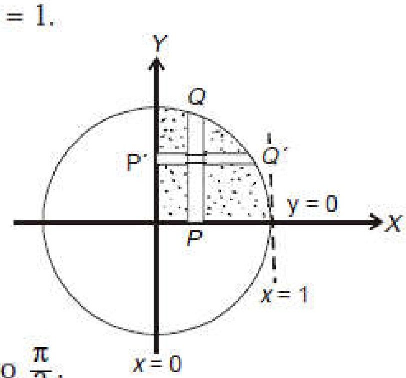
Example 1.

Evaluate the integral $\int_0^1\int_0^{\sqrt{1-x^2}} y^2 dydx$ by changing the order of integration.

Solution. In the above integral, y on vertical strip (say PQ) varies as a function of x and then the strip slides between $x = 0$ to $x = 1$.

Here $y = 0$ is the x-axis and $y=\sqrt{1-x^2}$ *i.e.*, $x^2 + y^2 = 1$ is the circle.

In the changed order, the strip becomes $P'Q'$, P' resting on the curve $x = 0$, Q' on the circle $x=\sqrt{1-y^2}$ and finally the strip $P'Q'$ sliding between $y = 0$ to $y = 1$.



$$\therefore \quad I = \int_0^1 y^2\left(\int_0^{\sqrt{1-y^2}} dx\right)dy$$

$$I = \int_0^1 y^2 [x]_0^{\sqrt{1-y^2}} dy$$

$$I = \int_0^1 y^2(1-y^2)^{\frac{1}{2}} dx$$

Substitute $y = \sin\theta$, so that $dy = \cos\theta\, d\theta$ and θ varies from 0 to $\frac{\pi}{2}$.

$$I = \int_0^{\frac{\pi}{2}} \sin^2\theta\cos^2\theta\, d\theta$$

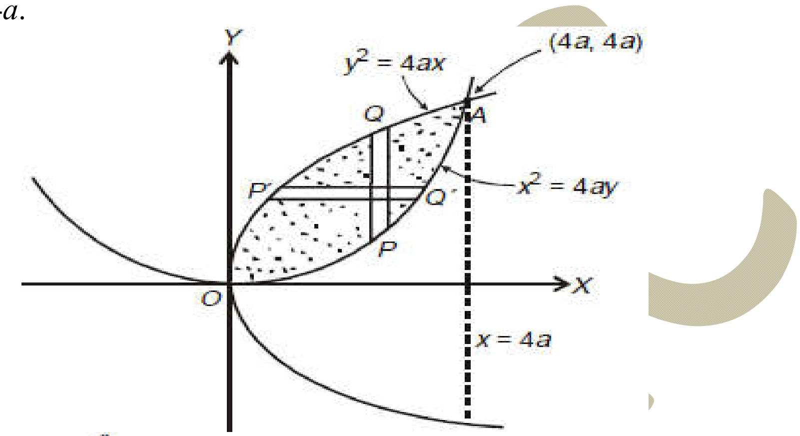
$$= \frac{(2-1)\cdot(2-1)}{4\cdot 2}\frac{\pi}{2} = \frac{\pi}{16}. \quad \underline{\text{Ans.}}$$

$$\left[\because \int_0^{\frac{\pi}{2}} \sin^p\theta\cos\theta\,d\theta = \frac{(p-1)(p-3)\ldots(q-1)(q-3)}{(p+q)(p+q-2)\ldots\ldots}\times\frac{\pi}{2},\ \text{only if both } p \text{ and } q \text{ are + ve even integers}\right]$$

Example 2.

Evaluate $\int_0^{4a}\int_{\frac{x^2}{4a}}^{2\sqrt{ax}} dydx$ by changing the order of integration.

Solution. In the given integral, over the vertical strip PQ (say), if y changes as a function of x such that P lies on the curve $y=\dfrac{x^2}{4a}$ and Q lies on the curve $y = 2\sqrt{ax}$ and finally the strip slides between $x = 0$ to $x = 4a$.



Here the curve $y=\dfrac{x^2}{4a}$ *i.e.* $x^2 = 4ay$ is a parabola with

$$\left.\begin{array}{lll} y=0 & \text{implying} & x=0 \\ y=4a & \text{implying} & x=\pm 4a \end{array}\right|$$

i.e., it passes through (0, 0) (4a, 4a), (– 4a, 4a).

Likewise, the curve $y=2\sqrt{ax}$ or $y^2 = 4ax$ is also a parabola with

$$x=0 \Rightarrow y=0 \text{ and } x=4a \Rightarrow y=\pm 4a$$

i.e., it passes through (0, 0), (4a, 4a), (4a, – 4a).

Clearly the two curves are bounded at (0, 0) and (4a, 4a).

$\therefore$ On changing the order of integration over the strip $P'Q'$, x changes as a function of y such that P' lies on the curve $y^2 = 4ax$ and Q' lies on the curve $x^2 = 4ay$ and finally $P'Q'$ slides between $y = 0$ to $y = 4a$.

whence
$$I=\int_0^{4a}\left(\int_{x=\frac{y^2}{4a}}^{x=2\sqrt{ay}} dx\right)dy$$

$$=\int_0^{4a}[x]_{\frac{y^2}{4a}}^{2\sqrt{ay}}\,dy$$

$$=\int_0^{4a}\left(2\sqrt{ay}-\frac{y^2}{4a}\right)dy$$

$$=\left[2\sqrt{a}\,\frac{y^{\frac{3}{2}}}{\frac{3}{2}}-\frac{y^3}{12a}\right]_0^{4a}=\frac{4\sqrt{a}}{3}(4a)^{\frac{3}{2}}-\frac{1}{12a}(4a)^3$$

$$=\frac{32a^2}{3}-\frac{16a^2}{3}=\frac{16a^2}{3}.$$

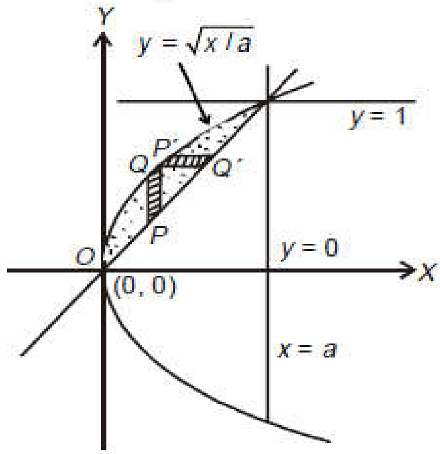
Ans.

Example 3.

Evaluate $\int_0^a \int_{\frac{x}{a}}^{\sqrt{\frac{x}{a}}} (x^2 + y^2)\, dx\, dy$ by changing the order of integration.

Solution.

In the given integral $\int_0^a \int_{x/a}^{\sqrt{x/a}} (x^2 + a^2)\, dx\, dy$, y varies along vertical strip PQ as a function of x and finally x as an independent variable varies from $x = 0$ to $x = a$.



Here $y = x/a$ i.e. $x = ay$ is a straight line and $y = \sqrt{x/a}$, i.e. $x = ay^2$ is a parabola.

For $x = ay$; $x = 0 \Rightarrow y = 0$ and $x = a \Rightarrow y = 1$.

Means the straight line passes through (0, 0), (a, 1).

For $x = ay^2$; $x = 0 \Rightarrow y = 0$ and $x = a \Rightarrow y = \pm 1$.

Means the parabola passes through (0, 0), (a, 1), (a, – 1),.

Further, the two curves $x = ay$ and $x = ay^2$ intersect at common points (0, 0) and (a, 1).

On changing the order of integration,

$$\int_0^a \int_{x/a}^{\sqrt{x/a}} (x^2 + y^2)\, dx dy = \int_{y=0}^{y=1} \left(\int_{x=ay^2}^{x=ay} (x^2 + y^2)\, dx dy \right)$$

(at P')

$$I = \int_0^1 \left[\frac{x^3}{3} + xy^2 \right]_{ay^2}^{ay} dy$$

$$= \int_0^1 \left[\left(\frac{(ay)^3}{3} + ay.y^2 \right) - \left(\frac{1}{3}(ay^2)^3 + ay^2 \cdot y^2 \right) \right] dy$$

$$= \int_0^1 \left[\left(\frac{a^3}{3} + a \right) y^3 - \frac{a^3}{3} y^6 - ay^4 \right] dy$$

$$= \left\{ \left(\frac{a^3}{3} + a \right) \frac{y^4}{4} - \frac{a^3}{3} \frac{y^7}{7} - \frac{ay^5}{5} \right\}_0^1$$

$$= \left\{ \left(\frac{a^3}{3 \times 4} - \frac{a^3}{3 \times 7} \right) + \left(\frac{a}{4} - \frac{a}{5} \right) \right\}$$

$$= \frac{a^3}{28} + \frac{a}{20} = \frac{a}{140}(5a^2 + 7).$$

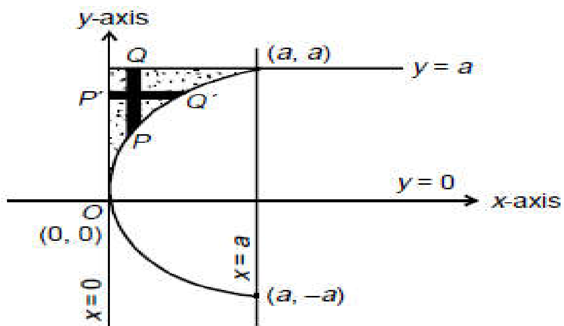
Ans.

Example 4. Evaluate $\int_0^a \int_{\sqrt{ax}}^a \frac{y^2}{\sqrt{y^4 - a^2x^2}}\, dydx$ by changing order of integration.

Solution. In the above integral I (say), y on the vertical strip (say PQ) varies as a function of x and then the strip slides between $x = 0$ to $x = a$.

Here the curve $y = \sqrt{ax}$ *i.e.*, $y^2 = ax$ is the parabola and the curve $y = a$ is the straight line.

On the parabola, $x = 0 \Rightarrow y = 0$; $x = a \Rightarrow y = \pm a$ *i.e.*, the parabola passes through points $(0, 0)$, (a, a) and $(a, -a)$.



Changing the order of integration, we have

$$I = \int_0^a \left(\int_{\substack{x=0 \\ (\text{at } P')}}^{x=\frac{y^2}{a}} \frac{y^2}{\sqrt{y^4 - a^2x^2}}\, dx \right) dy$$

$$= \int_0^a \left(\int_0^{\frac{y^2}{a}} \frac{y^2}{a} \frac{1}{\sqrt{\left(\frac{y^2}{a}\right)^2 - x^2}}\, dx \right) dy$$

$$= \int_0^a \frac{y^2}{a} \left[\sin^{-1} \frac{x}{\left(\frac{y^2}{a}\right)} \right]_0^{\frac{y^2}{a}} dy$$

$$= \int_0^a \frac{y^2}{a} \left[\sin^{-1} 1 - \sin^{-1} 0 \right] dy$$

$$= \int_0^a \frac{y^2}{a} \frac{\pi}{2}\, dy = \frac{\pi}{2a} \left. \frac{y^3}{3} \right|_0^a = \frac{\pi a^2}{6}.$$ <u>Ans.</u>

Example 5.

Change the order of integration of $\int_0^1 \int_{x^2}^{2-x} xy\, dy\, dx$ and hence evaluate the same.

Solution.

In the given integral $\int_0^1 \left(\int_{x^2}^{2-x} xydy \right) dx$, on the vertical strip PQ(say), y varies as a function of x and finally x as an independent variable, varies from 0 to 1.

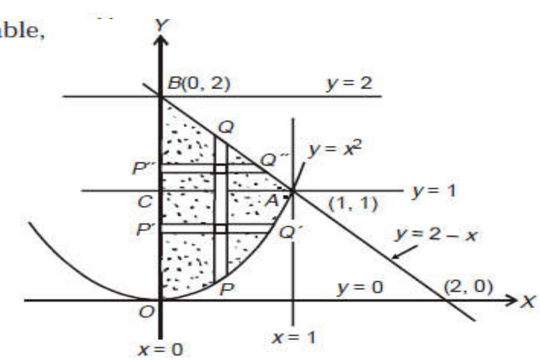
Here the curve $y = x^2$ is a parabola with

$$\left.\begin{array}{lll} y = 0 & \text{implying} & x = 0 \\ y = 1 & \text{implying} & x = \pm 1 \end{array}\right\}$$

i.e., it passes through (0, 0), (1, 1), (– 1, 1).

Likewise, the curve $y = 2 - x$ is straight line with

$$\left.\begin{array}{l} y = 0 \Rightarrow x = 2 \\ y = 1 \Rightarrow x = 1 \\ y = 2 \Rightarrow x = 0 \end{array}\right\}$$



i.e. it passes through (1, 1), (2, 0) and (0, 2).

On changing the order integration, the area $OABO$ is divided into two parts $OACO$ and $ABCA$. In the area $OACO$, on the strip $P'Q'$, x changes as a function of y from $x = 0$ to $x = \sqrt{y}$. Finally y goes from $y = 0$ to $y = 1$.

Likewise in the area ABCA, over the strip p"Q", x changes as a function of y from $x = 0$ to $x = 2 - y$ and finally the strip P"Q" slides between $y = 1$ to $y = 2$.

$$\therefore \quad \int_0^1\left(\int_0^{\sqrt{y}} xy\,dx\right)dy + \int_1^2\left(\int_0^{2-y} xy\,dx\right)dy$$

$$= \int_0^1\left(y\frac{x^2}{2}\Big|_0^{\sqrt{y}}\right)dy + \int_1^2\left(y\frac{x^2}{2}\Big|_0^{2-y}\right)dy$$

$$= \int_0^1 \frac{y^2}{2}\,dy + \int_1^2 \frac{y(2-y)^2}{2}\,dy$$

$$= \frac{1}{6} + \frac{1}{2}\left(2y^2 - \frac{4y^3}{3} + \frac{y^4}{4}\right)_1^2$$

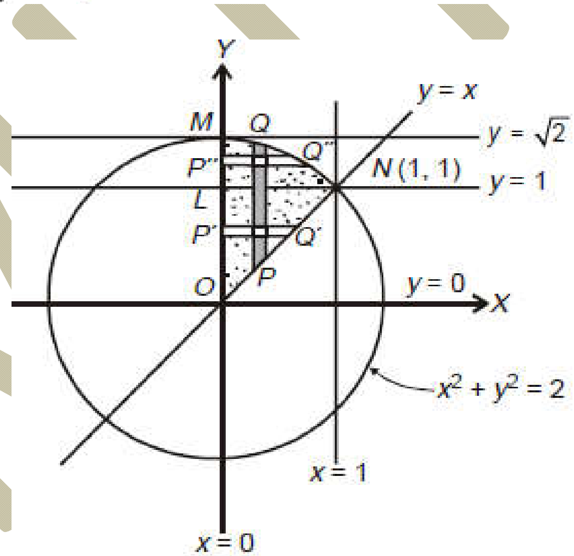
$$I = \frac{1}{6} + \frac{5}{24} = \frac{3}{8}.$$

Ans.

Example 6.

Evaluate $\int_0^1 \int_x^{\sqrt{2-x^2}} \frac{x}{\sqrt{x^2+y^2}}\,dy\,dx$ by changing order of integration.

Solution.



Clearly over the strip PQ, y varies as a function of x such that P lies on the curve $y = x$ and Q lies on the curve $y = \sqrt{2 - x^2}$ and PQ slides between ordinates $x = 0$ and $x = 1$.

The curve are $y = x$, straight line and $y = \sqrt{2 - x^2}$, i.e. $x^2 + y^2 = 2$, a circle. The common points of intersection of the two curves are (0, 0) and (1, 1).

On changing the order of integration, the same region $ONMO$ is divided into two parts $ONLO$ and $LNML$ with horizontal strips $P'Q'$ and $P''Q''$ sliding between $y = 0$ to $y = 1$ and $y = 1$ to $y = \sqrt{2}$ respectively.

whence $$I=\int_0^1\int_0^y \frac{x}{\sqrt{x^2+y^2}}\,dx\,dy+\int_1^{\sqrt{2}}\int_0^{\sqrt{2-y^2}}\frac{x}{\sqrt{x^2+y^2}}\,dx\,dy$$

Now the exp. $\dfrac{x}{x^2+y^2}=\dfrac{d}{dx}(x^2+y^2)^{\frac{1}{2}}$

$\therefore$ $$I=\int_0^1\left[(x^2+y^2)^{\frac{1}{2}}\right]_0^y dy+\int_1^{\sqrt{2}}\left[(x^2+y^2)^{\frac{1}{2}}\right]_0^{\sqrt{2-y^2}} dy$$

$$I=\int_0^1\left[(x^2+y^2)^{\frac{1}{2}}\right]_0^y dy+\int_1^{\sqrt{2}}\left[(x^2+y^2)^{\frac{1}{2}}\right]_0^{\sqrt{2-y^2}} dy$$

$$=(\sqrt{2}-1)\frac{y^2}{2}\Big|_0^1+\left(\sqrt{2}y-\frac{y^2}{2}\right)_0^{\sqrt{2}}=\frac{1}{2}(\sqrt{2}-1)$$ Ans.

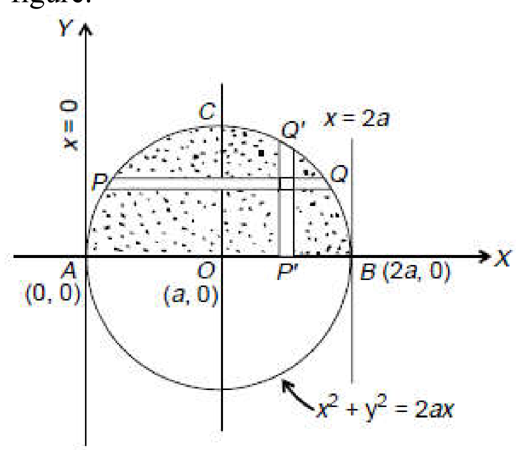
Example 7.

Evaluate $\displaystyle\int_0^a\int_{a-\sqrt{a^2-y^2}}^{a+\sqrt{a^2-y^2}} dy\,dx$ by changing the order of integration.

Solution.

Given $\displaystyle\int_{y=o}^{y=a}\left(\int_{x=a-\sqrt{a^2-y^2}}^{x=a+\sqrt{a^2-y^2}} dx\right)dy$

Clearly in the region under consideration, strip PQ is horizontal with point P lying on the curve $x=a-\sqrt{a^2-y^2}$ and point Q lying on the curve $x=a+\sqrt{a^2-y^2}$ and finally this strip slides between two abscissa $y = 0$ and $y = a$ as shown in below figure:



Now, for changing the order of integration, the region of integration under consideration is same but this time the strip is $P'Q'$ (vertical) which is a function of x with extremities P' and Q' at $y = 0$ and $y=\sqrt{2ax-x^2}$ respectively and slides between $x = 0$ and $x = 2a$.

Thus $$I=\int_0^{2a}\left(\int_0^{\sqrt{2ax-x^2}} dy\right)dx=\int_0^{2a}\left[y\right]_0^{\sqrt{2ax-x^2}} dx$$

$$=\int_0^{2a}\sqrt{2ax-x^2}\,dx=\int_0^{2a}\sqrt{x}\sqrt{2a-x}\,dx$$

Take $\sqrt{x}=\sqrt{2a}\sin\theta$ so that $dx=4a\sin\theta\cos\theta\,d\theta$,

Also, For $x=0$, $\theta=0$ and for $x=2a$, $\theta=\dfrac{\pi}{2}$

Therefore, $$I=\int_0^{\frac{\pi}{2}}\sqrt{2a}\sin\theta\cdot\sqrt{2a-2a\sin^2\theta}\cdot 4a\sin\theta\cdot\cos\theta\,d\theta$$

$$=8a^2\int_0^{\frac{\pi}{2}}\sin^2\theta\cos^2\theta\,d\theta=8a^2\cdot\frac{(2-1)(2-1)}{4(4-2)}\frac{\pi}{2}=\frac{\pi a^2}{2}$$ Ans.

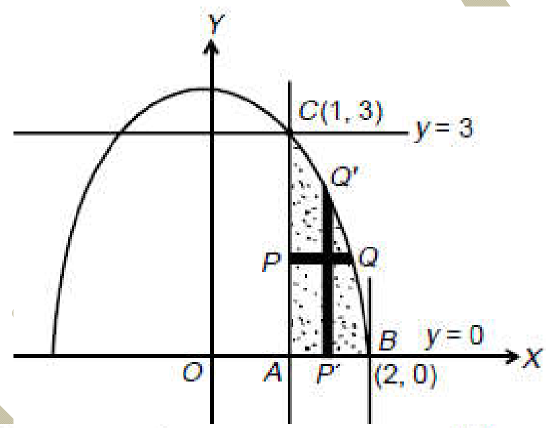
$$\left(\text{using} \int_0^{\frac{\pi}{2}} \sin^p\theta \cos^q\theta \, d\theta = \frac{(p-1)(p-3)\ldots(q-1)(q-3)\ldots}{(p+q)(p+q-2)\ldots\ldots\ldots\ldots\ldots\ldots} \frac{\pi}{2}, \quad p \text{ and } q \text{ both positive even integers}\right)$$

Example 8.

Changing the order of integration, evaluate $\int_0^3 \int_1^{\sqrt{4-y}} (x+y)\,dx\,dy$.

Solution. Clearly in the given form of integral, x changes as a function of y and y as an independent variable changes from 0 to 3. Thus the two curves are the straight line x = 1 and the parabola $x = \sqrt{4-y}$ and the common area under consideration is ABQCA.

For changing the order of integration, we need to convert the horizontal strip PQ to a vertical strip P'Q' over which y changes as a function of x and it slides for values of $x = 1$ to $x = 2$ as shown in figure below:



$$\therefore \quad I = \int_1^2 \left(\int_0^{(4-x^2)} (x+y)\,dy \right) dx = \int_1^2 \left[xy + \frac{y^2}{2} \right]_0^{4-x^2} dx$$

$$= \int_1^2 \left[x(4-x^2) + \frac{(4-x^2)^2}{2} \right] dx$$

$$= \int_1^2 \left[x(4-x^2) + \left(8 + \frac{x^4}{2} - 4x^2 \right) \right] dx$$

$$= \left[2x^2 - \frac{x^4}{4} + 8x + \frac{x^5}{10} - \frac{4}{3}x^3 \right]_1^2$$

$$= 2(2^2 - 1^2) - \frac{1}{4}(2^4 - 1^4) + 8(2-1) + \frac{1}{10}(2^5 - 1^5) - \frac{4}{3}(2^3 - 1^3)$$

$$= 6 - \frac{15}{4} + 8 + \frac{31}{10} - \frac{28}{3} = \frac{241}{60}.$$

Ans.

Change of Order of Integration in Double Integrals (Polar)

In the integral $\int_{\theta=\alpha}^{\theta=\beta}\int_{r=\phi(\theta)}^{r=\Psi(\theta)} f(r,\theta)\,dr\,d\theta$, interation is first performed with respect to r along a 'radial strip' and then this trip slides between two values of $\theta = \alpha$ to $\theta = \beta$.

In the changed order, integration is first performed with respect to θ (as a function of r along a 'circular arc') keeping r constant and then integrate the resulting integral with respect to r between two values $r = a$ to $r = b$ (say)

Mathematically expressed as

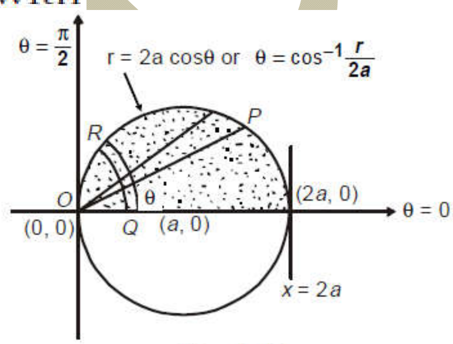
$$\int_{\theta=\alpha}^{\theta=\beta}\int_{r=\phi(\theta)}^{r=\Psi(\theta)} f(r,\theta)\,dr\,d\theta = I = \int_{r=a}^{r=b}\int_{\theta=f(r)}^{\theta=\eta(r)} f(r,\theta)\,d\theta\,dr$$

EXAMPLE 9.

Change the order of integration in the integral $\int_0^{\pi/2}\int_0^{2a\cos\theta} f(r,\theta)\,dr\,d\theta$

SOLUTION

Here, integration is first performed with respect to r (as a function of θ) along a **radial strip** OP (say) from $r = 0$ to $r = 2a\cos\theta$ and finally this radial strip slides between $\theta = 0$ to $\theta = \frac{\pi}{2}$.



Curve $\quad r = 2a\cos\theta \Rightarrow r^2 = 2ar\cos\theta$

$\Rightarrow \quad x^2 + y^2 = 2ax \Rightarrow (x-a)^2 + y^2 = a^2$

i.e., it is circle with centre $(a, 0)$ and radius a.

On changing the order of integration, we have to first integrate with respect to θ (as a function of r) along the '**circular strip**' QR (say) with pt. Q on the curve $\theta = 0$ and pt. R on the curve $\theta = \cos^{-1}\frac{r}{2a}$ and finally r varies from 0 to $2a$.

$$\therefore \qquad I = \int_0^{\frac{\pi}{2}}\int_0^{2a\cos\theta} f(r,\theta)\,dr\,d\theta = \int_0^{2a}\left(\int_0^{\cos^{-1}\frac{r}{2a}} f(r,\theta)\,d\theta\right)dr$$

EXAMPLE 10.

Sketch the region of integration $\int_a^{ae^{\frac{\pi}{4}}}\int_{2\log\frac{r}{a}}^{\pi/2} f(r,\theta)\,r\,dr\,d\theta$ and change the order of integration.

SOLUTION

Double integral $\int_0^{ae^{\pi/4}}\int_{2\log\frac{r}{a}}^{\pi/2} f(r,\theta)\,r\,dr\,d\theta$ is identical to $\int_{r=\alpha}^{r=\beta}\int_{\theta=f_1(r)}^{\theta=f_2(r)} f(r,\theta)\,r\,dr\,d\theta$, whence integration is first performed with respect to θ as a function of r *i.e.*, $\theta = f(r)$ along the 'circular strip' PQ (say) with point P on the curve $\theta = 2\log\frac{r}{a}$ and point Q on the curve $\theta = \frac{\pi}{2}$ and finally this strip slides between $r = a$ to $r = ae^{\pi/4}$. See Figure 1.

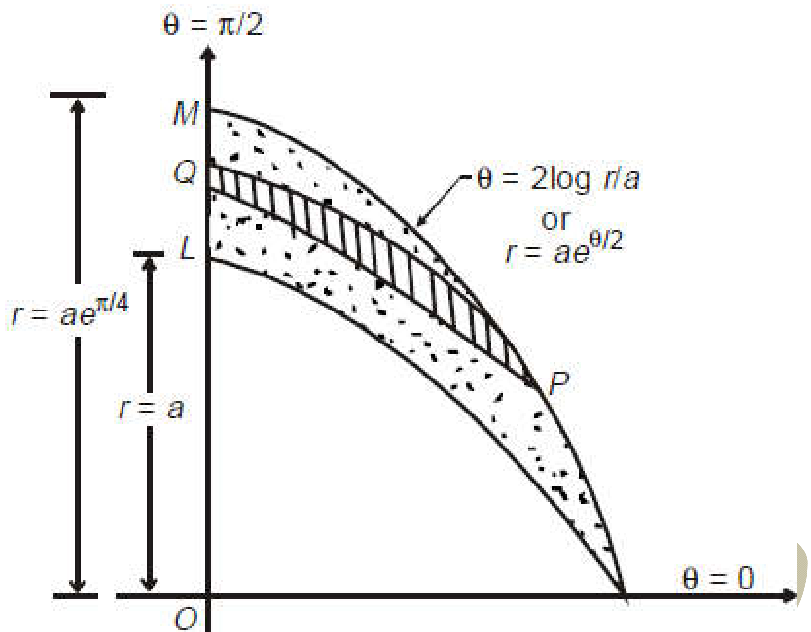


Figure 1

Now on changing the order, the integration is first performed with respect to r as a function of θ viz. $r = f(\theta)$ along the 'radial strip' PQ (say) and finally this strip slides between $\theta = 0$ to $\theta = \frac{\pi}{2}$. See Figure 2.

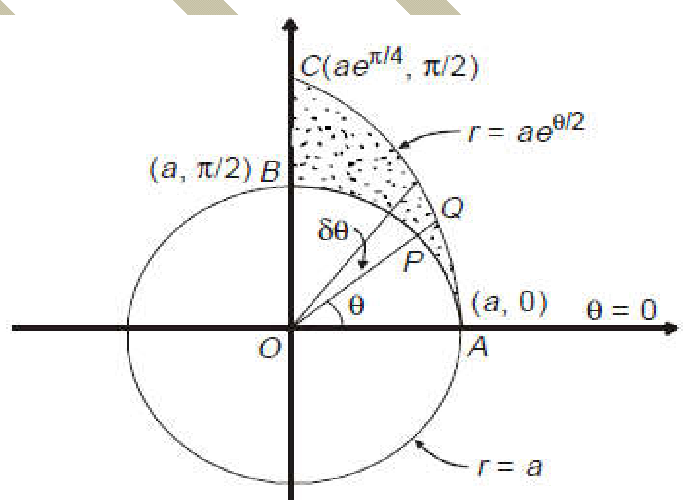


Fig. 2

$$\therefore \qquad I = \int_{\theta=0}^{\pi/2}\left(\int_{r=a}^{r=ae^{\theta/2}} f(r,\theta)\, r\, dr\right) d\theta$$

Figure 2